Learning From Bitcoin Fog Case

4/29/2024

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Overview

- Defendant *Roman Sterlingov* is charged in United States District Court, District of Columbia, with money laundering conspiracy, money laundering, operating an unlicensed money transmitting business, and money transmission without a license, all in relation to his alleged operation of a bitcoin mixer known as *Bitcoin Fog*.
- The defense challenged the reliability and validity of the *Chainalysis Reactor* software used by the prosecution, but the judge was satisfied that the evidence was the product of reliable principles and methods, thus denying the defense's request to exclude all testimony and evidence based on the software.
- A federal jury in Washington, D.C., convicted *Sterlingov* of all of the above. A federal district court judge will determine any sentence after considering the U.S. Sentencing Guidelines and other statutory factors.

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Parties

Prosecutor

- **United States of America**
  - **IRS-CI District of Columbia Cyber Crime Unit**
  - **FBI Washington Field Office**

Attorney

- *Jeff Pearlman* and *C. Alden Pelker* of the Criminal Division's Computer Crime and Intellectual Property Section (CCIPS)
- Assistant U.S. Attorney *Christopher B. Brown* for the D.C

Assistance

- CCIPS Paralegal Specialist *Divya Ramjee* and *Angela De Falco*
- Europool
- Swedish Economic Crime Authority
- Swedish Prosecution Authority
- Swedish Police Authority
- General Inspectorate of Romanian Police
- Directorate for Combating Organized Crime
- Directorate for Investigating Organized Crime and Terrorism
- **Chainalysis**

Defendant

- ***Roman Sterlingov***

Attorney

- *Tor Ekeland* and *Michael Hassard* of Tor Ekeland Law PLLC, in NY
- *Marina Medvin* of Medvin Law PLC, in VA

Assistance

- **CipherTrace**

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Chainalysis Reactor

- Chainalysis Reactor is a software product used to cluster cryptocurrency addresses that are likely controlled by the same entity and to then tie those clusters to particular entities based on information gleaned from other sources, including by conducting test transactions with those entities, researching open sources, and exchanging information with various cryptocurrency exchanges and law enforcement agencies.
- Chainalysis uses three types of heuristics "computational functions that rank different search algorithms at each branching step based on available information to decide which brand to follow."
  - Heuristic 1: co-spend or common spend heuristic

    Each transaction input requires that the sender have access to the private key for each of the corresponding input addresses, which means that it is very likely that a single person or entity controls each of the input addresses. This heuristic clusters several co- spending input addresses together under the assumption. "CoinJoin" services enable different individuals to contribute inputs to a single transaction, thereby defeating the assumption. But Chainalysis has controls to detect CoinJoin and can skip CoinJoin co-spends.
  - Heuristic 2: pattern recognition heuristic

    The heuristic is based on observing and tracking a particular entity's on-chain behaviors and patterns. Once particular behaviors have been identified, an algorithm can be used to cluster the potentially thousands of addresses that engage in transactions that match the pattern. For example, software wallets have distinctive ways of handling fees and change addresses," permitting Chainalysis to "investigate a service's particular transaction patterns" and to "develop clustering algorithms specific to that service."
  - Heuristic 3: intelligence-based heuristic

    It refers, instead, to information that Chainalysis has gathered off-chain, from sources such as "data leaks, court documents, Chainalysis data partnerships, exchanges that share their addresses with Chainalysis, and manual merges due to services changing wallets." Unlike Heuristics 1 and 2, which analyze the blockchain, this heuristic relies on information obtained from sources unrelated to any on-chain activity or analysis.
- For Bitcoin Fog, Reactor relied on Heuristic 1 (50.26%) and Heuristic 2 (49.74%) to cluster addresses.

link: https://caselaw.findlaw.com/court/us-dis-crt-dis-col/115886148.html

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Legal Standards

Daubert v. Merrell Dow Pharmaceuticals, Inc., 509 U.S. 579, 113 S.Ct. 2786, 125 L.Ed.2d 469 (1993)

- In the case, the Supreme Court identified certain factors that can inform the reliability analysis under Rule 702. Those factors include:

  a. whether the expert's theory or technique "can be (and has been) tested."
  b. whether it has been "subjected to peer review and publication"
  c. its "known or potential" error rate; and
  d. whether it has attracted widespread acceptance within a relevant scientific community

- The Supreme Court explained in another case that the test of reliability is 'flexible,' and Daubert's list of specific factors neither necessarily nor exclusively applies to all experts or in every case

- Ultimately, the trial court must assess the reliability of the expert testimony at issue based on "the particular circumstances of the particular case" and should apply, or decline to apply, the specific Daubert factors "depending on the nature of the issue."

Federal Rule of Evidence 702

- A district court has "broad discretion in determining whether to admit or exclude expert testimony.

- A qualified expert may testify if the "proponent demonstrates to the court that it is more likely than not that:"

  a. the expert's scientific, technical, or other specialized knowledge will help the trier of fact to understand the evidence or to determine a fact in issue;

  b. the testimony is based on sufficient facts or data;

  c. the testimony is the product of reliable principles and methods; and

  d. the expert's opinion reflects a reliable application of the principles and methods to the facts of the case.

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Defense Expert Report by CypherTrace

- The defense was assisted by experts from CypeherTrace. CipherTrace's involement was prompted by the defense attorneys for Sterlingov, and they provided a detailed 41-page report challenging the data and conclusions presented by Chainalysis, which was used by the prosecution.

- The summary of the report is as follows:
  - Questionable Reliability: A lack of independent verification and peer-review of the data and methodologies used by Chainalysis, raising doubts about their accuracy and reliability
  - Methodological Concerns: concerns around the use of certain heuristics by Chainalysis, particularly Heuristic 2, which the report described as vague and potentially over-inclusive / concerns around a high rate of false negatives, falsely linking innocent addresses to illicit activities
  - Discrepancies with CypherTrace: significant discrepancies when comparing Chainalysis' attributions with those from CipherTrace, which suggests inconsistencies in how addresses were clustered and attributed to Bitcoin Fog
  - Lack of Transparency: a lack of transparency in Chainalysis' processes, with no detailed explanation provided on how specific subcategories of Heuristic 2 were used and how they impacted the data's accuracy.

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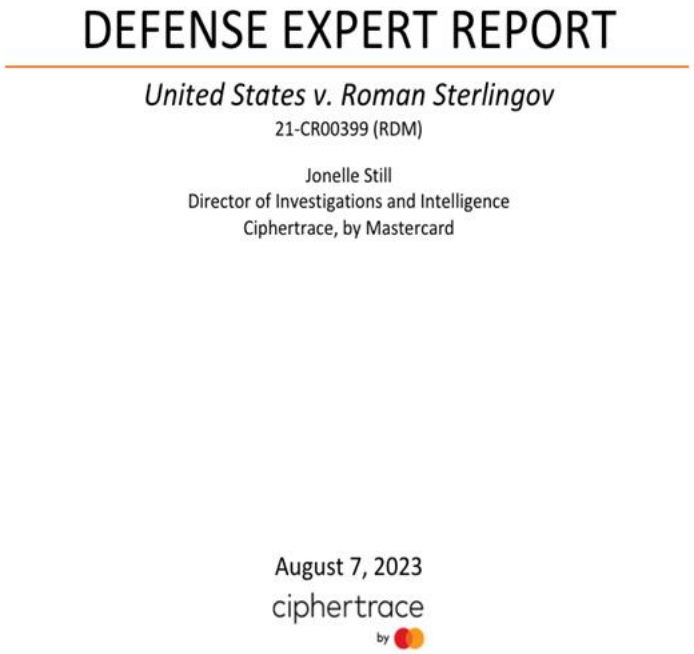

DEFENSE EXPERT REPORT

United States v. Roman Sterlingov

21-CR00399 (RDM)

Jonelle Still
Director of Investigations and Intelligence
Ciphertrace, by Mastercard

August 7, 2023

ciphertrace by

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Court’s Response 1

- This is not a case in which the governemnt’s theory that Sterlingov was the operator of Bitcoin Fog turns exclusively, or even primarily, on Scholl and Bisbee’s use of Reactor software. Rather, in its effort to establish that crucial point, the government relies in substantial part on materials found in Sterlingov’s possession when he was arrested, various posts on an online forum called Bitcoin Talk, IP analyses. In the words of the defense, the testimony that will be offered based on the use of the Reactor software constitutes a “minor witness” in the case.
- In challenging the government's use of the Reactor software, the defense focuses exclusively on subsection (c) of Rule 702. When the Court inquired whether the defense wished to raise “any of the other 702 issues,” the defense did not clearly raise any.
- The defense, instead, maintains that the government's reliance on Reactor fails each of the four Daubert factors, which principally concern subsection (c) of Rule 702. However, in making the assessment of whether the testimony is based on “reliable principles and methods,” trial courts have substantial “latitude in deciding how to test an expert’s reliability” and in deciding “whether or not that expert’s testimony is reliable.” The Court finds that Reactors easily clears the threshold for reliability and admissibility.
- For all of these reasons and based on the extensive testimony and expert reports in this case, the Court is persuaded that blockchain analytics in general, and Reactor in particular, is not junk science. ome of the defense's arguments might (or might not) offer fruitful ground for cross examination before the jury. The Court's role, however, is to act only as a gatekeeper, and, applying a preponderance of the evidence standard, to ensure that the testimony offered for the jury's consideration is “the product of reliable principles and methods.

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Court's Response 2

Daubert factors	Court's Response
a) whether the expert's theory or technique "can be (and has been) tested."	Reactor's clustering can be and has been tested. Clustering, whether conducted by Chainalysis or any other blockchain analytics company, can be replicated by competitor software products - Scholl corroborated Reactor's clustering through manual tracing and through TRM's software. A difference in size between Chiphertrace's Bitcoin Fog cluster and Chainalysis's one does not matter a lot and they shared meaningful overlap.
b) whether it has been "subjected to peer review and publication"	Chainalysis Reactor has not itself been subject to peer review, but the co-spend heuristic has receive widespread academic approval. Heuristic 2 varies from case-to-case and entity-to-entity, which makes it neither surprising nor dispositive that the algorithms developed for the case have not been the subject of peer review. Also, as Daubert itself made clear "[t]echnical fields need not be held to the standard of peer review applicable to traditional sciences, which are often considered in scholarly journals."
c) its "known or potential" error rate	The lack of a compiled "error rate" does not alter the Court's finding that Reactor is reliable. As detailed above, Scholl offered persuasive testimony concerning the lack of false positives in his extensive experience using Reactor and as confirmed by the tracing he performed in this case. Even the clustering by Ciphertrace and testified to by Still confirmed much of the work done by Reactor's Heuristic.
d) whether it has attracted widespread acceptance within a relevant scientific community	[Blockchain analysis] has been used extensively by law enforcement in the United States, by law enforcement all around the world, by private sector, by financial institutions, by consulting firms, by incident response firms, by regulators.

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Data Analysis vs Forensic Analysis

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Data Analysis vs Forensic Analysis

- Each clustering technique combines multiple data types depending on the purpose or case.
- The use of techniques tailored to non-universal human behaviors, product characteristics, or specific cases, especially when utilizing data not publicly accessible, can be defined as forensic analysis (?)

Clustering Techniques	Data Type (1) on-chain data	Data Type (2) (2) Data identified through data leaks and undercover investigation	Data Type (3) (3) Data identified through governmental subpoena power	Data Type (4) (4) Data shared by entities
Clustering based on technical and scientific theories (e.g., co-spend clustering, multi-input clustering)	This is the primary data type used. Transaction histories and relationships between addresses are analyzed.	May be used as supplemental information (e.g., for verification or to provide additional context for clusters)	Typically not used, but legally obtained data might contribute to improving clustering accuracy in certain cases	Can be used when provided
Clustering based on the behavioral patterns of individuals or organizations and the characteristics of products.	Used to analyze how specific groups of addresses are being analyzed	May help identify behavioral patterns	Can be used to supplement details about behavioral patterns	Used when behavioral data is directly provided by clients or services
Clustering based on manual processes tailored to specific cases using data such as attribution	May not always be the primary data type but can be used depending on the case	Critical in specific case-based manual processes	Used to support manual approaches in specific cases	Frequently used for custom solutions tailored to specific cases

Data analysis vs Forensic analysis

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Recommendations

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Recognition in Academia

Q1, 3, 4, 8

- Only techniques based on technical and scientific theories have the potential to become 'widely discussed and relied upon in academia' (as they were used to satisfy Daubert factor a in the Bitcoin Fog case). However, it may be necessary to establish criteria for determining what qualifies as such.
- For techniques not recognized in academia, if they are deemed acceptable as long as they are supported by other evidence (e.g., undercover operations, anonymous reviews, or analyses using other software), it is essential to at least demonstrate that such supporting credible evidence has been collected. But how?

Clustering Techniques	Recognition in Academia
Clustering based on technical and scientific theories (e.g., co-spend clustering, multi-input clustering)	○
Clustering based on the behavioral patterns of individuals or organizations and the characteristics of products.	✕
Clustering based on manual processes tailored to specific cases using data such as attribution	✕

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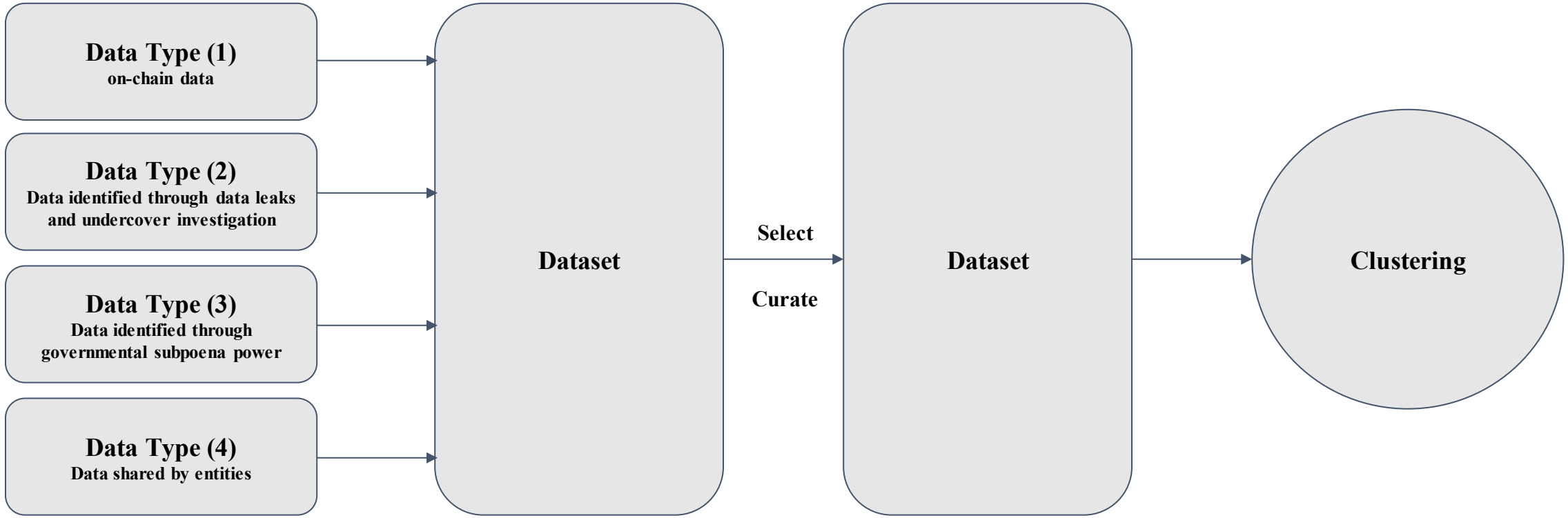
Data Access

Q 6

- The disclosure of the data used is crucial for ensuring data reliability and the reproducibility of analysis. However, considering that some of this data is confidential, blockchain analytics firms might need to at least give to their clients or the court an indication of which data types were used and how the data were hand-curated.



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Over-inclusivity and Error Rates

Q 2, 3, 4, 6, 7, 8

- The confidentiality of data must be preserved, and the analytical methods themselves often constitute the core competency of analysis firms. As these are not disclosed, it becomes difficult for external parties to identify issues such as over-inclusivity or errors. Therefore, using the inability of external parties to identify over-inclusivity or errors as a basis for asserting the reliability of these methods is not appropriate."
- Furthermore, if the reliability of these methods can be asserted solely based on undercover operations or reviews conducted by the analysis firms themselves, it opens the possibility for these firms to claim whatever they wish without sufficient external verification.
- While the data and techniques used by analysis firms remain a black box, the confirmation of meaningful overlap through analyses conducted using competing firms' software can serve as supporting evidence. However, it may be necessary to define what constitutes 'meaningful overlap.'

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Widespread Acceptance

Q 9

- A point of concern is that the frequent use of Chainalysis Reactor by government agencies, combined with testimony regarding the minimal occurrence of false positives by the U.S. government, has been accepted by the court as supporting evidence under Daubert factor d.
- This risks creating a situation where only analyses conducted by established large firms like Chainalysis are seen as meeting Daubert factor d. There should be clear and fair criteria established to allow smaller firms to meet the requirements of Daubert factor d as well.

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